

1.Air Filter to MAF Sensor:

Low Leak

Meaning: Minor intake bypass, no immediate performance impact

Recommendation:

1. Inspect air filter element for light dust loading.
2. Verify filter housing lid is seated correctly.
3. Continue test and schedule inspection during next planned stop.

Medium Leak

Meaning: Unfiltered air entering intake, long-term sensor risk

Recommendation:

1. Inspect air filter element for heavy dust or moisture.
2. Check housing seal and intake bellows for cracks or loose clips.
3. Verify if the service indicator has tripped.
4. Pause testing after current run and correct before continuation.

High Leak

Meaning: Severe filtration bypass, engine damage risk

Recommendation:

1. Stop test immediately.
2. Replace air filter element if contaminated or wet.
3. Repair or replace cracked housing / bellows.
4. Re-baseline MAF and Delta-P values before restart.

2. MAF Sensor to Turbo Compressor Inlet:

Low Leak

Meaning: Minor unmetered air, negligible turbo impact

Recommendation:

1. Monitor turbo inlet pressure trend.
2. Inspect hose clamps during scheduled downtime.
3. Test may continue.

Medium Leak

Meaning: Turbo efficiency reduced, data accuracy affected

Recommendation:

1. Inspect flexible hose for softening, collapse, or tears.
2. Check clamp torque to prevent unmetered air entry.
3. Inspect turbo inlet mesh for debris.
4. Resume testing only after corrective action.

High Leak

Meaning: Turbo under-speed, invalid test results

Recommendation:

1. Stop test immediately.
2. Replace inlet hose assembly.
3. Verify turbo inlet pressure stability.
4. Re-run baseline before resuming test cycle.

3. Turbo Compressor Outlet → Charge Air Cooler

Low Leak

Meaning: Minor boost loss

Recommendation:

1. Log pressure and temperature deviation.
2. Inspect boots during engine cooldown.
3. Continue test with monitoring.

Medium Leak

Meaning: Reduced charge density, power loss

Recommendation:

1. Inspect silicone boots for ballooning or pinhole leaks.
2. Check turbo outlet V-band clamp alignment.
3. Listen for whistling noise during acceleration.
4. Correct before extended testing.

High Leak

Meaning: Severe boost loss, turbo overspeed risk

Recommendation:

1. Stop test immediately.
2. Replace damaged boots or clamps.
3. Pressure test charge-air system.
4. Validate boost pressure before restart.

4. Charge Air Cooler → Intake Manifold (CS4)

Low Leak

Meaning: Slight MAP deviation

Recommendation:

1. Monitor MAP/MAT trends.
2. Inspect during next maintenance window.
3. Test may proceed.

Medium Leak

Meaning: Poor combustion efficiency

Recommendation:

1. Perform smoke test on CAC core.
2. Inspect intake manifold gasket and O-rings for oil/air weeping.
3. Check MAT sensor for oil fouling.
4. Resume testing after repair.

High Leak

Meaning: Severe boost loss, combustion instability

Recommendation:

1. Immediate shutdown.
2. Repair or replace CAC or manifold seals.
3. Clean or replace fouled sensors.
4. Revalidate intake pressure and temperature baseline.

5. Cylinder → Turbo Turbine Inlet (HS1)

Low Leak

Meaning: Minor exhaust energy loss

Recommendation:

1. Monitor EGT and turbo speed.
2. Continue testing with caution.

Medium Leak

Meaning: Turbo response degraded

Recommendation:

1. Inspect exhaust manifold gaskets for soot ghosting.
2. Check for heat cracks near center cylinders.
3. Retorque manifold fasteners.
4. Correct before high-load testing.

High Leak

Meaning: Hot gas escape – safety hazard

Recommendation:

1. Stop test immediately.
2. Replace cracked manifold or failed gasket.
3. Inspect nearby components for heat damage.
4. Restart only after full exhaust integrity check.

6. Turbine Outlet → Tailpipe (Aftertreatment Path)

DOC Section

- **Low:** Continue test; emissions trend deviation logged
- **Medium:** Inspect inlet flange for leaks affecting light-off
- **High:** Stop test; repair flange to prevent catalyst damage

DPF Section

- **Low:** Monitor Delta-P trend
- **Medium:** Inspect Delta-P sensor pipes for leaks
- **High:** Immediate repair; incorrect soot loading risk

SCR Section

- **Low:** Minor NOx deviation, log event
- **Medium:** Inspect DEF injector seal
- **High:** Stop test; visible crystallization indicates severe leak

7. System-Level Recommendation (Supervisory Logic)

Leak Severity: LOW

Condition

- Leak detected but engine operation stable

Recommendation

- Continue testing with monitoring enabled
- Log deviation as early warning event

Purpose

Allow uninterrupted testing while tracking leak progression.

Leak Severity: MEDIUM**Condition**

- Sustained deviation from baseline models

Recommendation

- Complete current test cycle
- Schedule corrective action before next test phase

Purpose

Maintain data validity and prevent escalation.

Leak Severity: HIGH**Condition**

- Critical leak detected in any intake or exhaust zone

Recommendation

- Suspend performance and emissions testing immediately
- Rectify identified leak
- Revalidate steady-state sensor baselines before resuming testing

Purpose

Protect hardware, ensure test integrity, and prevent aftertreatment damage.

8. Healthy State (No Leak Detected)**Normal Condition****Recommendation:**

1. Baseline performance confirmed.
2. Log current sensor values as the **Golden Reference**.
3. Proceed to next test stage or clear engine for shipment.
4. No mechanical intervention required.